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Field combinations 2 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	constant	$F\ D$	$S_1\ D$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$		$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1 + i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2$		$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0		$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1 + i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2$		$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$		$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1 + i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2$		$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1		$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1 + i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2$		$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$		$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1 + i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2$		$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$		$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 + i \overline{F_{r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1$		
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0		$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 + \frac{1}{2} i F_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$		$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 + i \overline{F_{r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1$		

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	constant	$F\ D$	$S_1\ D$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger\right)^2 \lambda_{S_2}^1$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} \left(F_{j_1,r_1}\right)$	
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-M_{S_2}^2 S_2 S_2^\dagger$ $+S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger\right)^2 \lambda_{S_2}^1$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} \left(F_{j_1,r_1}\right)$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger\right)^2 \lambda_{S_2}^1$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} \left(F_{j_1,r_1}\right)$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-M_{S_2}^2 S_2 S_2^\dagger$ $+S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger\right)^2 \lambda_{S_2}^1$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} \left(F_{j_1,r_1}\right)$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger\right)^2 \lambda_{S_2}^1$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} \left(F_{j_1,r_1}\right)$	
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$		$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} \left(F_{j_1,r_1}\right)$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0		$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} \left(F_{j_1,r_1}\right)$	
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$		
									$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	constant	$F\ D$	$S_1\ D$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2 \end{aligned}$	$+i\overline{F_{j_1,r_1}}\gamma^{\mu 1}D_{\mu 1}\left(F_{j_1,r_1}\right)$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2 \end{aligned}$	$+i\overline{F_{j_1,r_1}}\gamma^{\mu 1}D_{\mu 1}\left(F_{j_1,r_1}\right)$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2 \end{aligned}$	$+i\overline{F_{j_1,r_1}}\gamma^{\mu 1}D_{\mu 1}\left(F_{j_1,r_1}\right)$	

Field combinations 3 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	$S_1 H$	$S_2 D$	$S_2 F$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\begin{aligned} & + \overline{L_{a_1, i_1}} C P_R \overline{F_{j_1, r_1}}^T S_1 y_{1, i_1, r_1} \mathcal{I}_{\mathcal{Y}^1, a_1, j_1} \\ & + \overline{y_{1, i_1, r_1}} L_{a_1, i_1}^T C P_L F_{j_1, r_1} \text{Bar}(\text{T3Y1CG})_{a_1, j_1} S_1^\dagger \end{aligned}$	$\begin{aligned} & + \overline{\lambda_{\text{T3}} S_{2, j_1}} \text{Bar}(\text{T3MixCG})_{a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger \\ & + H_{a_1} H_{a_2} S_1 \mathcal{I}_{\text{T3}, a_1, a_2, j_1} \lambda_{\text{T3}} S_{2, j_1}^\dagger \\ & + H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{\text{H1}, a_1, a_2}^1 \lambda_{\text{H1}}^1 \end{aligned}$		
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\begin{aligned} & + \overline{L_{a_1, i_1}} C P_R \overline{F_{j_1, r_1}}^T S_1 y_{1, i_1, r_1} \mathcal{I}_{\mathcal{Y}^1, a_1, j_1} \\ & + \overline{y_{1, i_1, r_1}} L_{a_1, i_1}^T C P_L F_{j_1, r_1} \text{Bar}(\text{T3Y1CG})_{a_1, j_1} S_1^\dagger \end{aligned}$	$\begin{aligned} & + \overline{\lambda_{\text{T3}} S_{2, j_1}} \text{Bar}(\text{T3MixCG})_{a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger \\ & + H_{a_1} H_{a_2} S_1 \mathcal{I}_{\text{T3}, a_1, a_2, j_1} \lambda_{\text{T3}} S_{2, j_1}^\dagger \\ & + H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{\text{H1}, a_1, a_2}^1 \lambda_{\text{H1}}^1 \end{aligned}$		
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & + \overline{L_{a_1, i_1}} C P_R \overline{F_{j_1, r_1}}^T S_1 y_{1, i_1, r_1} \mathcal{I}_{\mathcal{Y}^1, a_1, j_1} \\ & + \overline{y_{1, i_1, r_1}} L_{a_1, i_1}^T C P_L F_{j_1, r_1} \text{Bar}(\text{T3Y1CG})_{a_1, j_1} S_1^\dagger \end{aligned}$	$\begin{aligned} & + \overline{\lambda_{\text{T3}} S_{2, j_1}} \text{Bar}(\text{T3MixCG})_{a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger \\ & + H_{a_1} H_{a_2} S_1 \mathcal{I}_{\text{T3}, a_1, a_2, j_1} \lambda_{\text{T3}} S_{2, j_1}^\dagger \\ & + H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{\text{H1}, a_1, a_2}^1 \lambda_{\text{H1}}^1 \end{aligned}$		
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\begin{aligned} & + \overline{L_{a_1, i_1}} C P_R \overline{F_{j_1, r_1}}^T S_1 y_{1, i_1, r_1} \mathcal{I}_{\mathcal{Y}^1, a_1, j_1} \\ & + \overline{y_{1, i_1, r_1}} L_{a_1, i_1}^T C P_L F_{j_1, r_1} \text{Bar}(\text{T3Y1CG})_{a_1, j_1} S_1^\dagger \end{aligned}$	$\begin{aligned} & + \overline{\lambda_{\text{T3}} S_{2, j_1}} \text{Bar}(\text{T3MixCG})_{a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger \\ & + H_{a_1} H_{a_2} S_1 \mathcal{I}_{\text{T3}, a_1, a_2, j_1} \lambda_{\text{T3}} S_{2, j_1}^\dagger \\ & + H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{\text{H1}, a_1, a_2}^1 \lambda_{\text{H1}}^1 \end{aligned}$		
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\begin{aligned} & + \overline{L_{a_1, i_1}} C P_R \overline{F_{j_1, r_1}}^T S_1 y_{1, i_1, r_1} \mathcal{I}_{\mathcal{Y}^1, a_1, j_1} \\ & + \overline{y_{1, i_1, r_1}} L_{a_1, i_1}^T C P_L F_{j_1, r_1} \text{Bar}(\text{T3Y1CG})_{a_1, j_1} S_1^\dagger \end{aligned}$	$\begin{aligned} & + \overline{\lambda_{\text{T3}} S_{2, j_1}} \text{Bar}(\text{T3MixCG})_{a_1, a_2, j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger \\ & + H_{a_1} H_{a_2} S_1 \mathcal{I}_{\text{T3}, a_1, a_2, j_1} \lambda_{\text{T3}} S_{2, j_1}^\dagger \\ & + H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{\text{H1}, a_1, a_2}^1 \lambda_{\text{H1}}^1 \end{aligned}$		
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$				
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0				
T3-B	0	2	0	2	1	1	$\frac{1}{2}$				
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1				
T3-B	2	2	1	2	2	1	$\frac{3}{2}$				
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$				
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0				
T3-C	0	2	0	2	1	3	$\frac{1}{2}$				
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1				
T3-C	2	2	1	2	2	3	$\frac{3}{2}$				
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$			$+ D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$\begin{aligned} & + \overline{y_{2, i_1, r_1} F_{j_1, r_1}} P_L L_{a_1, i_1} S_2 \text{Bar}(\text{T3Y2CG})_{a_1, j_1} \\ & + \overline{L_{a_1, i_1}} P_R F_{j_1, r_1} y_{2, i_1, r_1} \mathcal{I}_{\mathcal{Y}^2, a_1, j_1} S_2^\dagger \end{aligned}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0			$+ D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$\begin{aligned} & + \overline{y_{2, i_1, r_1} F_{j_1, r_1}} P_L L_{a_1, i_1} S_2 \text{Bar}(\text{T3Y2CG})_{a_1, j_1} \\ & + \overline{L_{a_1, i_1}} P_R F_{j_1, r_1} y_{2, i_1, r_1} \mathcal{I}_{\mathcal{Y}^2, a_1, j_1} S_2^\dagger \end{aligned}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$			$+ D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$\begin{aligned} & + \overline{y_{2, i_1, r_1} F_{j_1, r_1}} P_L L_{a_1, i_1} S_2 \text{Bar}(\text{T3Y2CG})_{a_1, j_1} \\ & + \overline{L_{a_1, i_1}} P_R F_{j_1, r_1} y_{2, i_1, r_1} \mathcal{I}_{\mathcal{Y}^2, a_1, j_1} S_2^\dagger \end{aligned}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1			$+ D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$\begin{aligned} & + \overline{y_{2, i_1, r_1} F_{j_1, r_1}} P_L L_{a_1, i_1} S_2 \text{Bar}(\text{T3Y2CG})_{a_1, j_1} \\ & + \overline{L_{a_1, i_1}} P_R F_{j_1, r_1} y_{2, i_1, r_1} \mathcal{I}_{\mathcal{Y}^2, a_1, j_1} S_2^\dagger \end{aligned}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$			$+ D_{\mu_1} (S_2)^\dagger D_{\mu_1} (S_2)$	$\begin{aligned} & + \overline{y_{2, i_1, r_1} F_{j_1, r_1}} P_L L_{a_1, i_1} S_2 \text{Bar}(\text{T3Y2CG})_{a_1, j_1} \\ & + \overline{L_{a_1, i_1}} P_R F_{j_1, r_1} y_{2, i_1, r_1} \mathcal{I}_{\mathcal{Y}^2, a_1, j_1} S_2^\dagger \end{aligned}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$				
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0				
T3-E	0	3	0	3	1	2	$\frac{1}{2}$				
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1				
T3-E	2	3	1	3	2	2	$\frac{3}{2}$				

Field combinations 4 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{L}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+ \overline{\lambda_{T3}}S_2\text{Bar}(T3\text{MixCG})_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+ H_{a_2}S_2H_{a_1}^\dagger S_2^\dagger \mathcal{L}_{H2,a_1,a_2}{}^1\lambda_{H2}{}^1$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{L}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+ \overline{\lambda_{T3}}S_2\text{Bar}(T3\text{MixCG})_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+ H_{a_2}S_2H_{a_1}^\dagger S_2^\dagger \mathcal{L}_{H2,a_1,a_2}{}^1\lambda_{H2}{}^1$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{L}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+ \overline{\lambda_{T3}}S_2\text{Bar}(T3\text{MixCG})_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+ H_{a_2}S_2H_{a_1}^\dagger S_2^\dagger \mathcal{L}_{H2,a_1,a_2}{}^1\lambda_{H2}{}^1$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{L}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+ \overline{\lambda_{T3}}S_2\text{Bar}(T3\text{MixCG})_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+ H_{a_2}S_2H_{a_1}^\dagger S_2^\dagger \mathcal{L}_{H2,a_1,a_2}{}^1\lambda_{H2}{}^1$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{L}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+ \overline{\lambda_{T3}}S_2\text{Bar}(T3\text{MixCG})_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+ H_{a_2}S_2H_{a_1}^\dagger S_2^\dagger \mathcal{L}_{H2,a_1,a_2}{}^1\lambda_{H2}{}^1$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	